Hello,
Nice to “meet” you

I’m Christine Wu

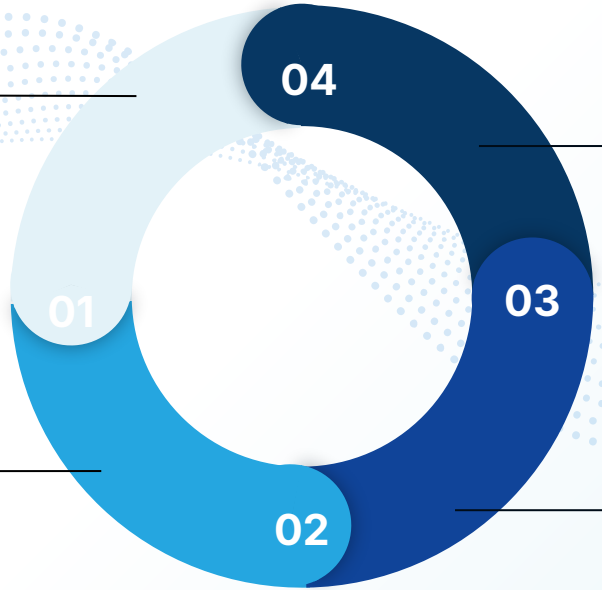
Agenda - Presentation & Additional

Selected Projects

- Electrohydrodynamic process (HER Catalyst & Li-ion battery)

Un-selected Projects

- MoS2 through CVD for Field Effect Transistor(FET)
- Hydrogen production
- Mithrene X-Ray examination



Extracurricular Activities

- Science fairs
- Leadership
- Volunteering

Interests & Hobbies

- Reading (philosophy & literature)
- Enjoy nature

Selected Project

3D Printing of “dewetting regulated organization of porous structure(s)” (DROPS) with Electrohydrodynamic Process

What would happen?



Flow chart of Electrohydrodynamic process



HER Catalyst Project

MoS2 Homostructure

Li-ion Battery Project

MoS2/rGO Heterostructure

Temperature

Electric field

Position

Solvent ratio

Vertical/Lateral directions

Composition ratio

mechanically robust, radiation tolerant, chemical resilience and selective patterns

Chemicals involved (MoS2 & rGO)

Bulk Molybdenum disulfide (MoS2)

Bulk 2H-MoS2 (hexagonal)

Bulk 3R-MoS2 (rhombohedral)

Intercalating with alkali metals (Li)

1T-MoS2 (tetragonal)

Graphite Oxide

Oxidizers

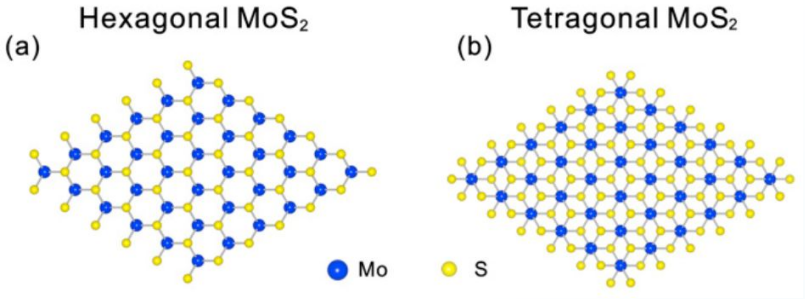
Graphite

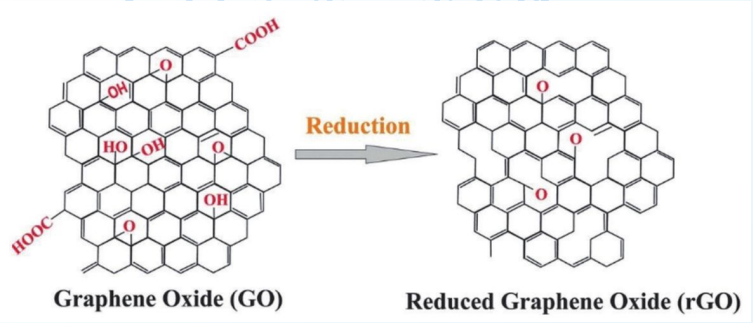
Chemical/Physical treatments

Graphene Oxide (GO)

Reduction

Reduced Graphene Oxide (rGO)

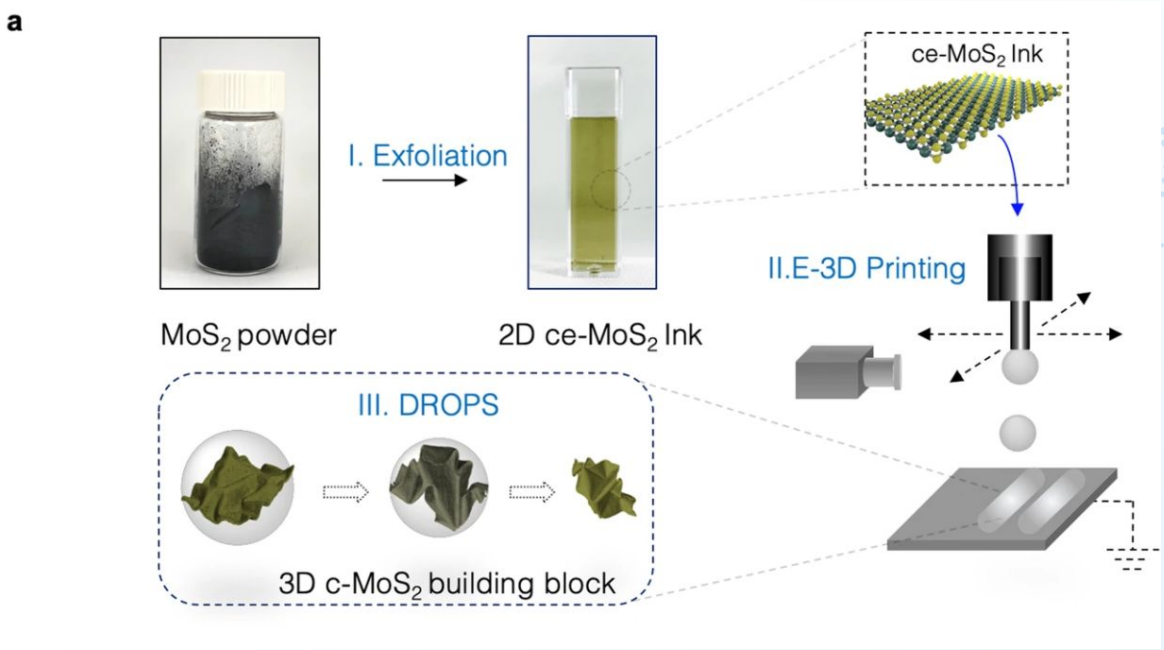




Three steps to create 3D homo- and - heterostructures

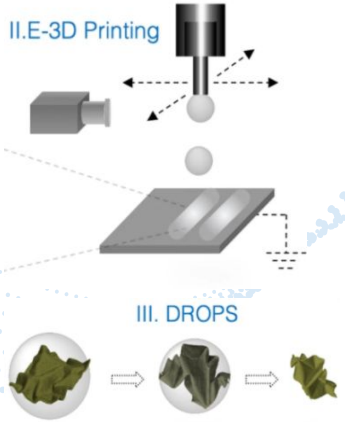
Steps:

I. Exfoliation
II. E-3D Printing
III. DROPS



Example of chemical process for MoS2 homostructures

Three boundary conditions for successful DROPS structures



1. MoS2 medium (liquid) does not wet the supporting substrates & remains a stable, nonplanar shape upon evaporation
2. MoS2 medium (liquid) is chemically nonreactive with the substrate

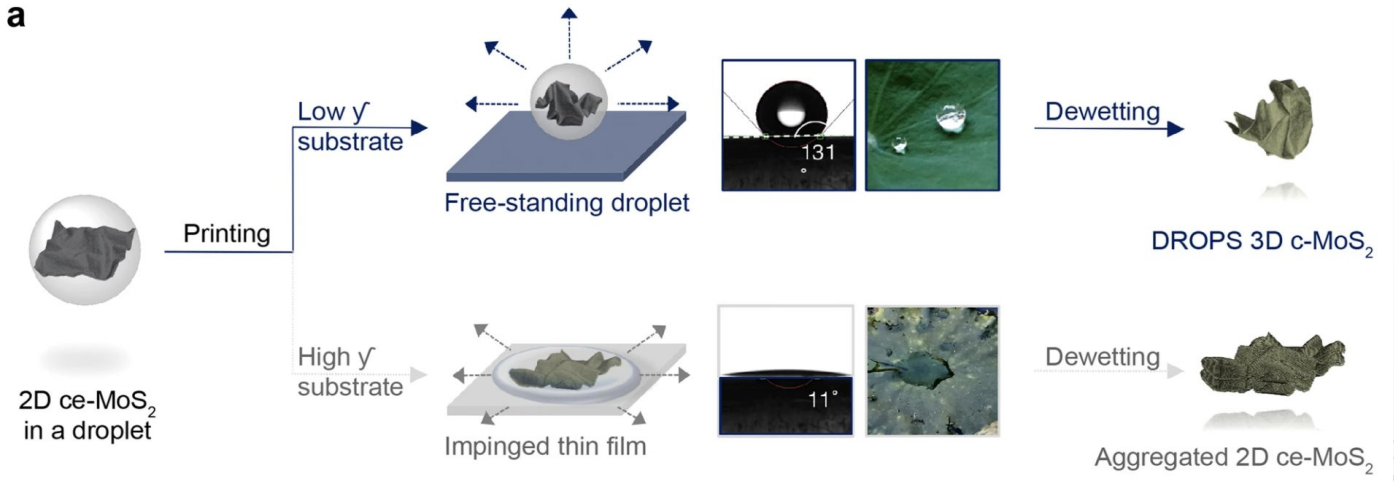
Interfacial energies between liquid/substrate

3. Printed patterns do not flatten into thin films upon continuous printing process

Work of adhesion between substrate/liquid in air

Note: Appendix A-I has detail descriptions

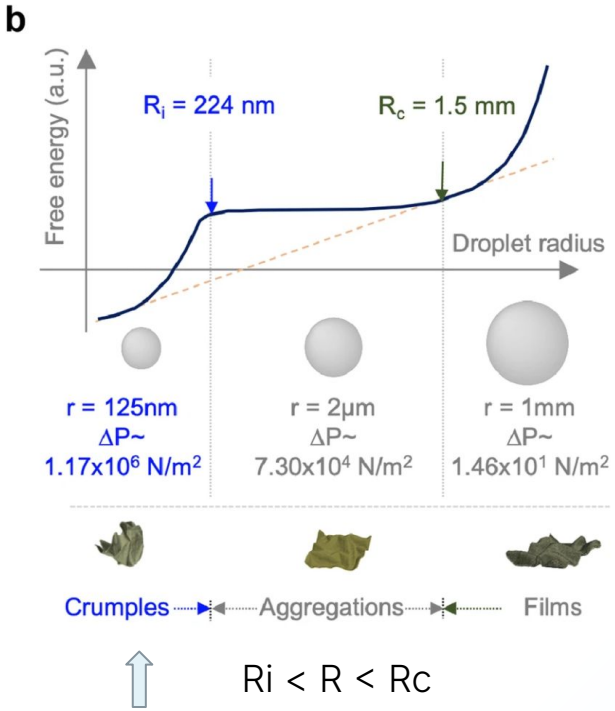
Dewetting process with interfacial energies & work of adhesion



Note: Appendix A-II ~ III has detail calculations

*Where 1 is substrate; 2 is liquid; 3 is air

Laplace pressures affects morphology of 3D-c-MoS2



Smaller r, Larger ΔP

$$Ri = (\kappa^{-1} a)^{1/2}$$

K^{-1} : capillary length
a: molecular length (for H2O = 0.275 nm)
Ri of 244 nm

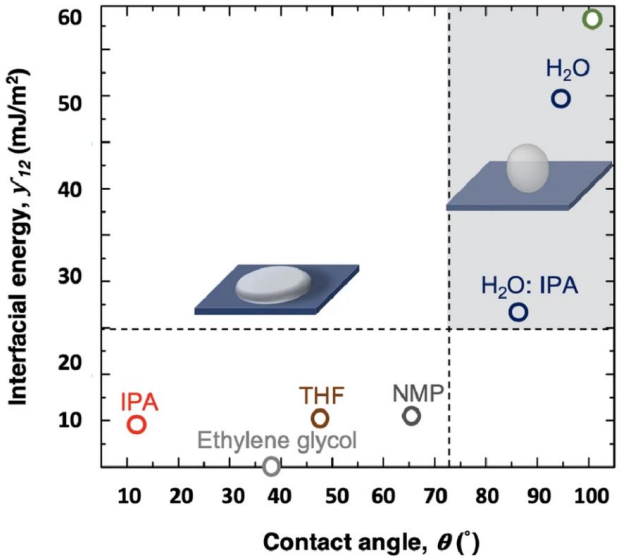
$$\Delta P = 2\gamma_{13}/r$$

γ13: interfacial energy substrate/air (73 mJ/m2) *
r: 125 nm (< Ri)
$\Delta P \approx 2 \times (73 \times 10\text{-}3)/125 \times 10\text{-}9 \approx 1.17 \times 10^6$ N/m^2

*Take example of HOPG substrate

Note: Appendix A-II ~ V has detail calculations

Contact angles of polar solvents as a function of interfacial energies



Based on quantitative descriptors:

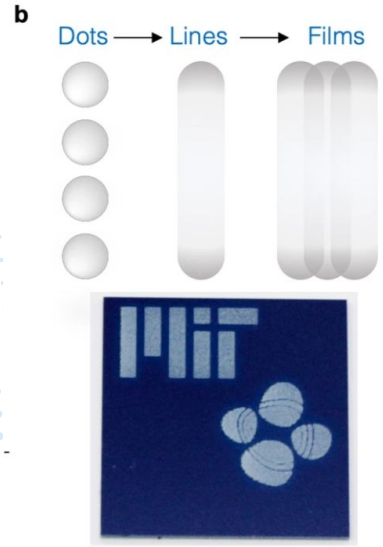
$$\gamma_{12} > 0, S < 0, and\ r < R_i$$

Fifter from data in supplementary table 1-2 in Appendix VI-VII

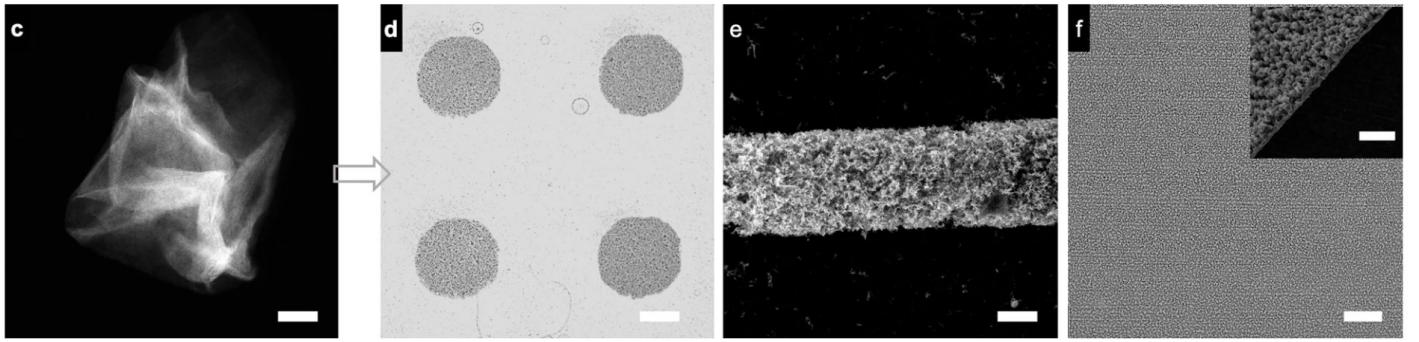
Solvent-substrate combinations for effectively driving DROPS where contact angle >85° and interfacial energy > 25 mJ/m^2

* Where γ12 is interfacial energy between substrate/liquid; S is spreading coefficient; r is radius, and Ri is inflection point radius

Versatility of E-3D printing (feature size, geometry & area coverage)



Dots Lines Films

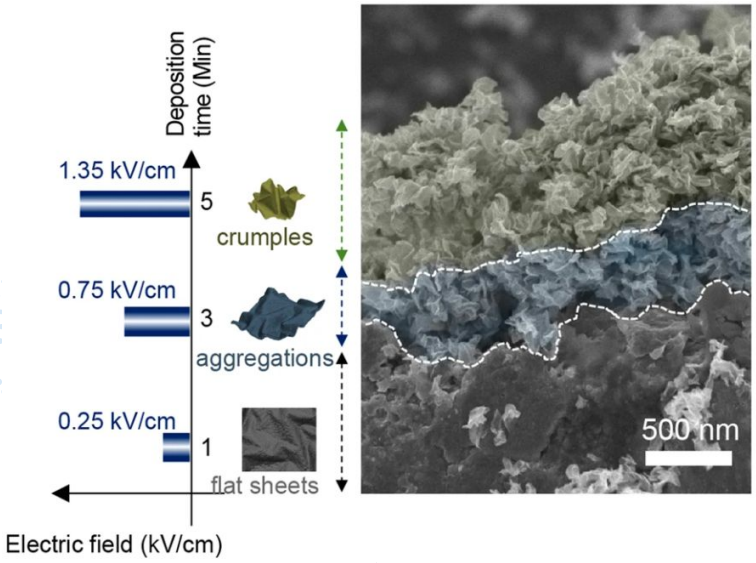


c) TEM image (25 nm) d) SEM image (1 µm) e) SEM image (5 µm) f) SEM image (100 µm)

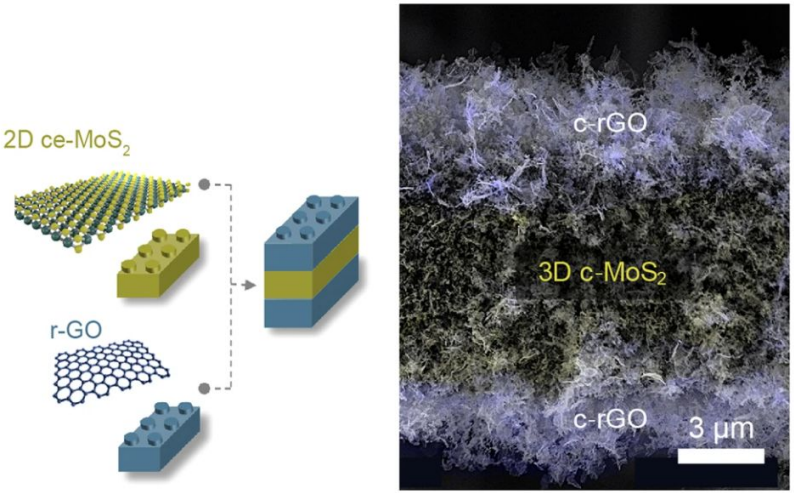
Propagated ridges, jagged facets, folded edges

Design freedom in structural complexity & compositional diversity

MoS2 homostructure



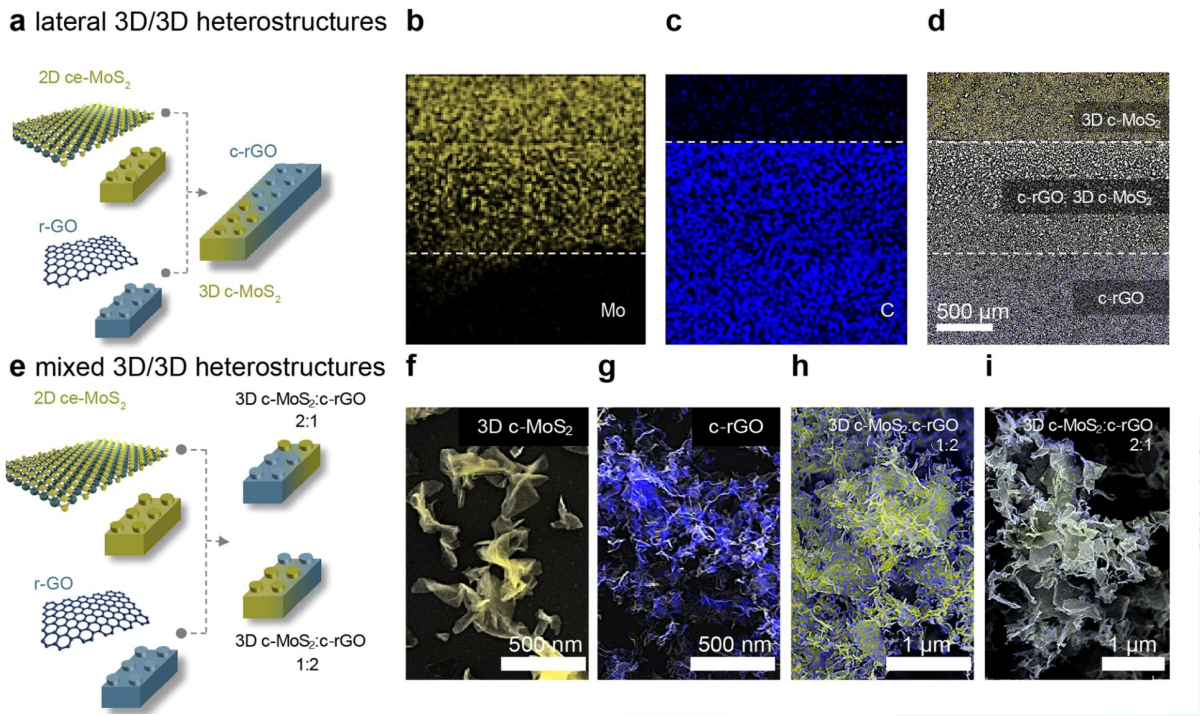
rGO & MoS2 heterostructure



Variations in electric field

Variations in compositions

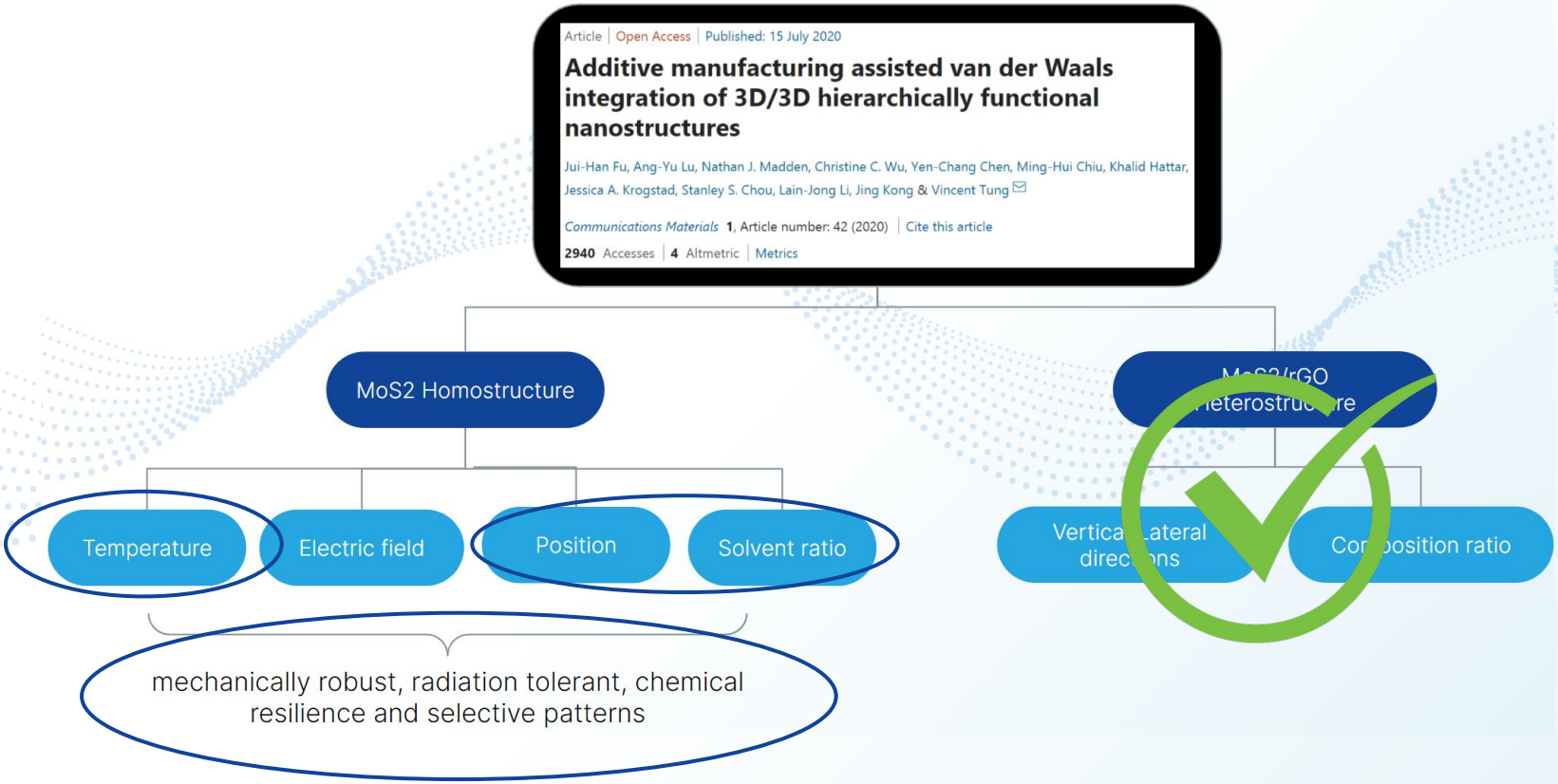
Heterostructure variations (lateral direction & composition ratio)



Variations in lateral direction

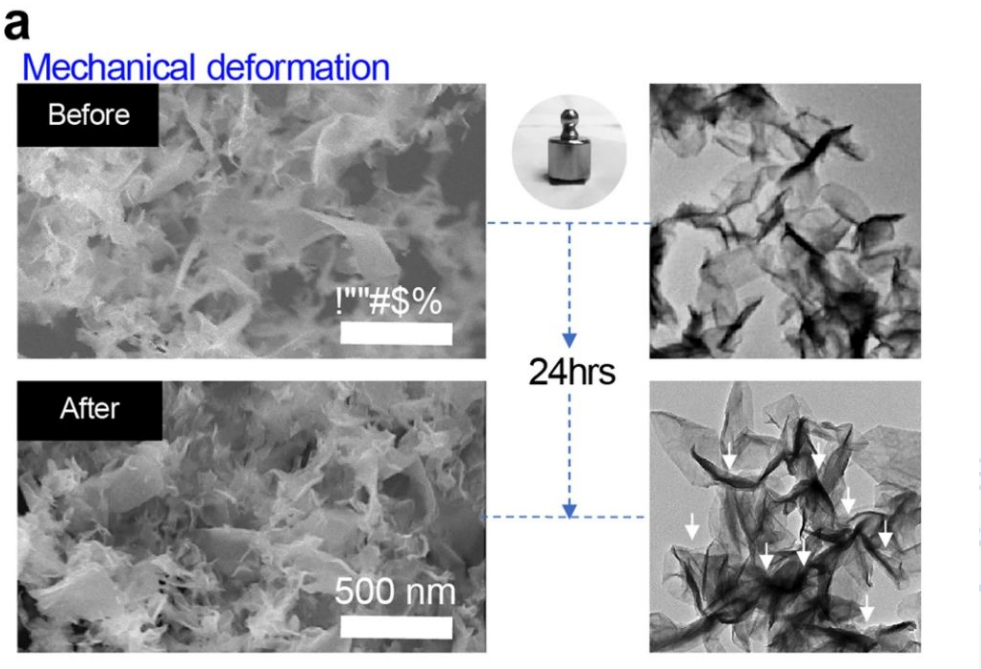
Variations in composition ratios

Flow chart of DROPS process revisit



MoS2 homostcure - Property I: Mechanically robust

Withstand external loads without catastrophic collapse



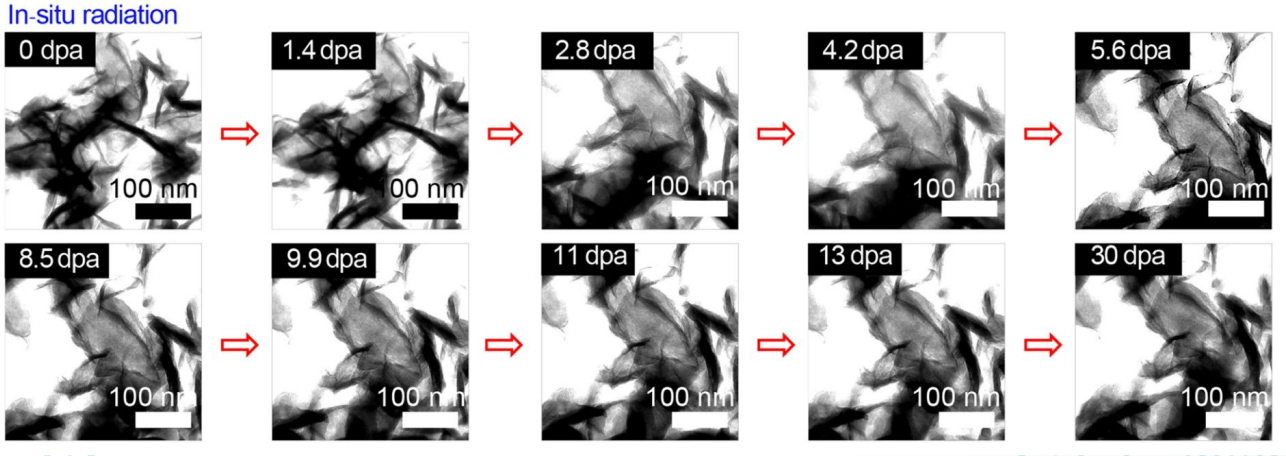
Remains intact after deformation for a duration of 24 hrs

Elastic buckling of extremely thin edges

The merging, scrolling, and elastic buckling of extremely thin and flexible edges from neighboring 3D c-MoS2 to give resilient struts

MoS2 homostructure - Property II: High radiation tolerance

In-situ ion irradiation TEM of 46 KeV Au^+ Bombardment during nanoscale observation



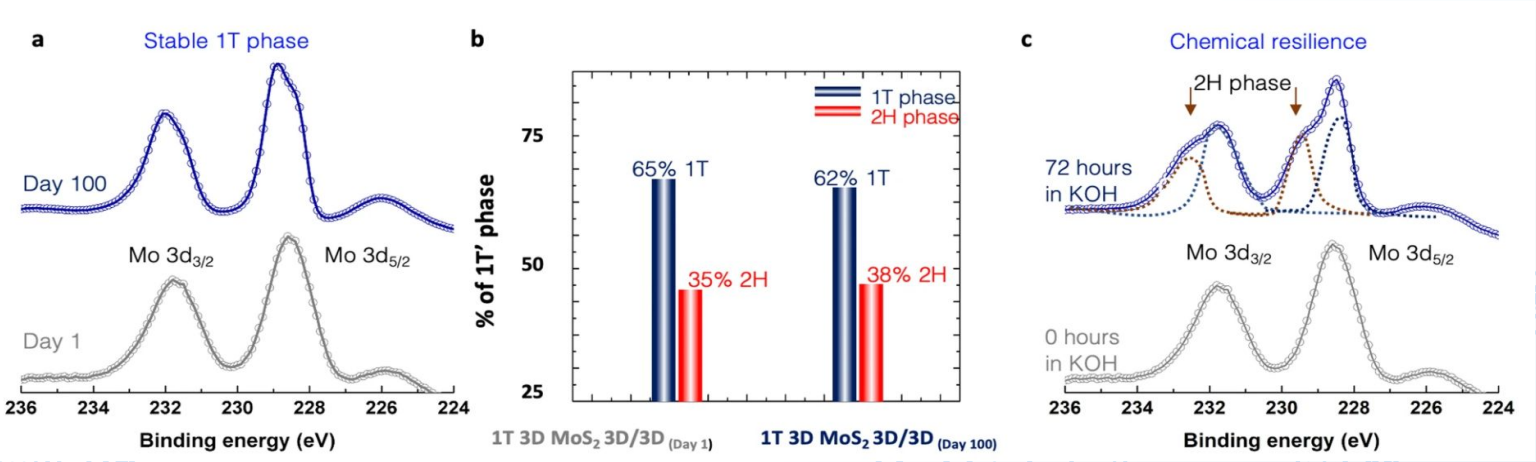
No significant structural degradation or distortion

Building blocks tightly bound at edges

Conditions:

Fluence (ions/cm^2)	Time (min)	Ion beam (keV)	Temperature	Displacement per atoms (dpa)
3×10^{15}	105	46 Au^+	Room Temp	30

MoS2 homostcure - Property III: Chemically resilient



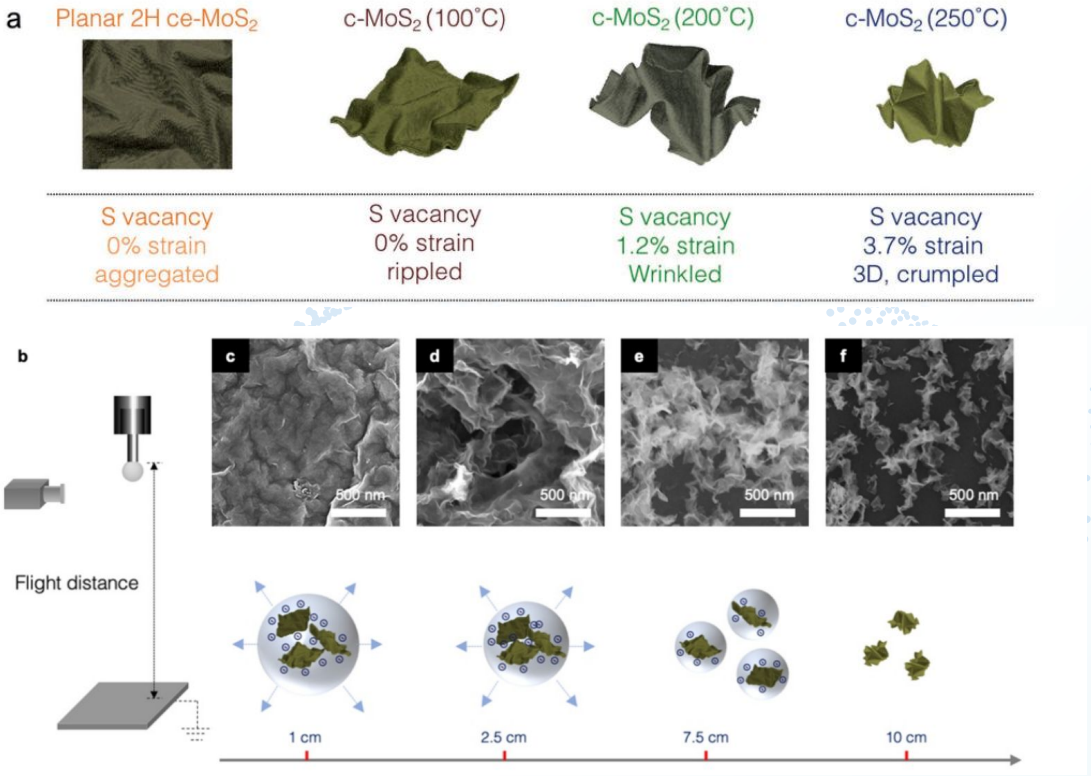
Core level photoemission peaks

3% degradation of 1T phase

Emergence of corrugated ridges

Overcome the metastable and temperature vulnerable nature of 1T phase, have concurrent retention of high concentration of 1 T metallic phase

Other variables - temperature & distance



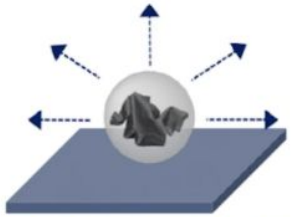
⇦ Strain increases as temperature increases (up to 250°C)
- HER Catalyst project

⇦ Proper distance with fission generation
- Analogy to Li-ion battery project

Future plan

- Generalization of materials (2D materials)
- Improve prototype systems (HER, Li-ion battery)
- Scaling up of process (manufacturing of automatic depositions for complex system)

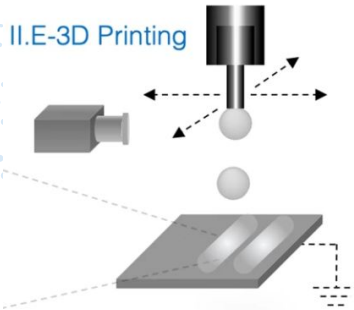
Appendix A-I: Interfacial energies comparison (Boundary I)



Free-standing droplet

⇦ γ_{12}

Substrate/liquid



⇦ γ_{23}

Liquid/air

⇦ γ_{13}

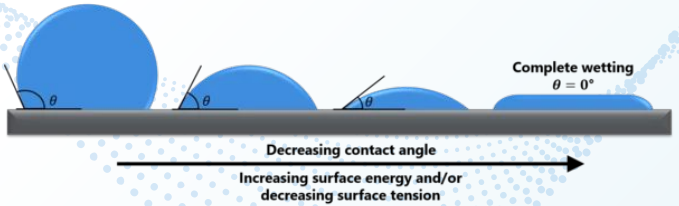
Substrate/air

*Where 1 is substrate; 2 is liquid; 3 is air

Interfacial Energy Comparison

$$\gamma_{12} = \gamma_{13} - \gamma_{23} cos\theta_0 > 0$$

θ_0 the equilibrium contact angle of the dispersing liquid and supporting substrate



- γ_{12} >0: substrate and liquid tend to separate from each other (phase separation or non-wetting)
- γ_{12} <0: interfaces between substrate and liquid will expand indefinitely, leading to dissolution or spreading

Appendix A-II: total surface energy & work of adhesion(Boundary III)

The final total surface energy of the entire system

$$W_{tot} = \gamma_{23}(A_C + A_f) - W_{132}A_f$$

*Where W_{132} represents the total energy required to cleave medium 1 from medium 2 within medium 3, A_C and A_f refer to curved and flat areas of the droplet

Insert equilibrium condition

$$\gamma_{23}(1 + cos\theta_0) = W_{132} = \gamma_{13} + \gamma_{23} - \gamma_{12}$$

New expression

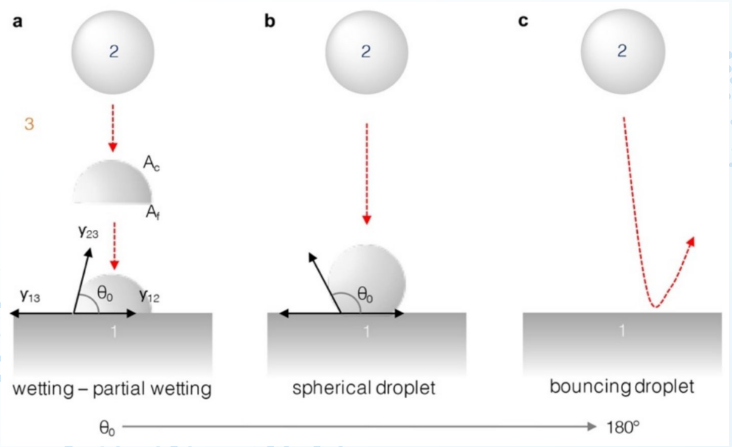
$$W_{tot} = [(A_C + A_f)/(1 + cos\theta_0) - A_f]W_{132}$$

In terms of radius

$$W_{tot} = 4\pi R_0^3 \gamma_{23}/R$$

*Where 1 is substrate; 2 is liquid; 3 is air

The correlation between contact angles, surface, and interfacial energies



- When W_{132} = 0, R equals R_0, W_{tot} remains unchanged. No adhesion and therefore no adhesive flattering will arise
- W_{tot} decrease as contact angle decreases and as R increases from R_0 to ∞. Droplets thus spread on the substrate

Appendix A-III: work of adhesion cont'd (Boundary III)

Hard to estimate the final radii of polydisperse droplets

⇩

$$W_{123} = W_{12} + W_{33} - W_{13} - W_{23}$$

$$W_{12} + W_{33} - W_{13} - W_{23} = \gamma_{13} + \gamma_{23} - \gamma_{12}$$

- When W132 > 0 : attraction between media 1 and 2 in medium 3
- When W132 < 0 : repulsion between media 1 and 2 in medium 3
  - ⇨ "Spreading coefficient" (S): medium 3 will displace medium 2
  - ⇨ Spread on or completely wet on the surface of medium 1
  - ⇨ Rending the spreading of droplets

*Where 1 is substrate; 2 is liquid; 3 is air

Appendix A-IV: Radius for initiating DROPS (non-wetting)

In a non-wetting regime (S<0), large droplets flatten out while smaller droplets remain stable or level out to form puddles

Ri < R < Rc

Inflection point (Ri)

$F''(R) = 0$

$(\kappa^{-1}a)^{1/2}$

$F(R) = \gamma_{12} + \gamma_2 + P(R) + 1/2\rho g R^2$

Where a is the molecular length of H2O » 0.275 nm.Thus obtain Ri of 244 nm

Critical radius (Rc)

$2\kappa^{-1}sin(\theta/2)$

where k^-1 and θ are capillary length and contact angle of H2O

Using k^-1 of 2.7 mm, and θ of 108˚ on CC substrate, Rc of » 1.5 mm.

*Where 1 is substrate; 2 is dispersing medium; 3 is air

Appendix A-V: Associated Laplace pressure (crumpling)

Upon evaporation, the interface between liquid/air is replaced by an amount of dR. The work done by the pressure and capillary forces

$$\delta W = -P_0 dV_0 - P_w dV_w + \gamma_{12} dA$$

$$dV_0 = 4\pi R^2 dR = -dV_w \qquad dA = 8\pi R dR$$

Laplace pressure: $\Delta P = 2\gamma_{13}/R$

For a droplet in stage 3, putting γ13 = 73 mJ/m2, average droplet radius of 125 nm (< Ri), we obtain ΔP ≈ 2 × (73 × 10-3)/125 × 10-9 ≈ 1.17 × 106 N/m2

*Where P_0 is the atmosphere pressure, Pw is the pressure inside the droplets
*Where 1 is substrate; 2 is liquid; 3 is air

Appendix A-VI: Supplementary table 1

Supplementary Table 1. Comparison of the governing parameters with theoretical prediction and experimental observations for various "*solid-liquid*" combinations.

Substrate	Medium	γ_1 (mJ/m^2)	γ_2 (mJ/m^2)	θ (°)	γ_{12} (mJ/m^2)	S (mJ/m^2)	Spherical droplet	
							Theory	Exp.
HOPG	H_2O	26	73	108	48.6	<0	Y	Y
HOPG	Glycerol	26	62.5	125	33.7	<0	Y	Y
HOPG	H_2O: IPA	26	27	103	32.7	<0	Y	Y
HOPG	NMP	26	40	72	13.6	≤0	Y/N	N
HOPG	THF	26	26.4	57	11.5	>0	Y/N	N
HOPG	IPA	26	21.7	7.5	9.1	>0	Y/N	N
HOPG	Ethylene glycol	26	48	40	-10	>0	N	N

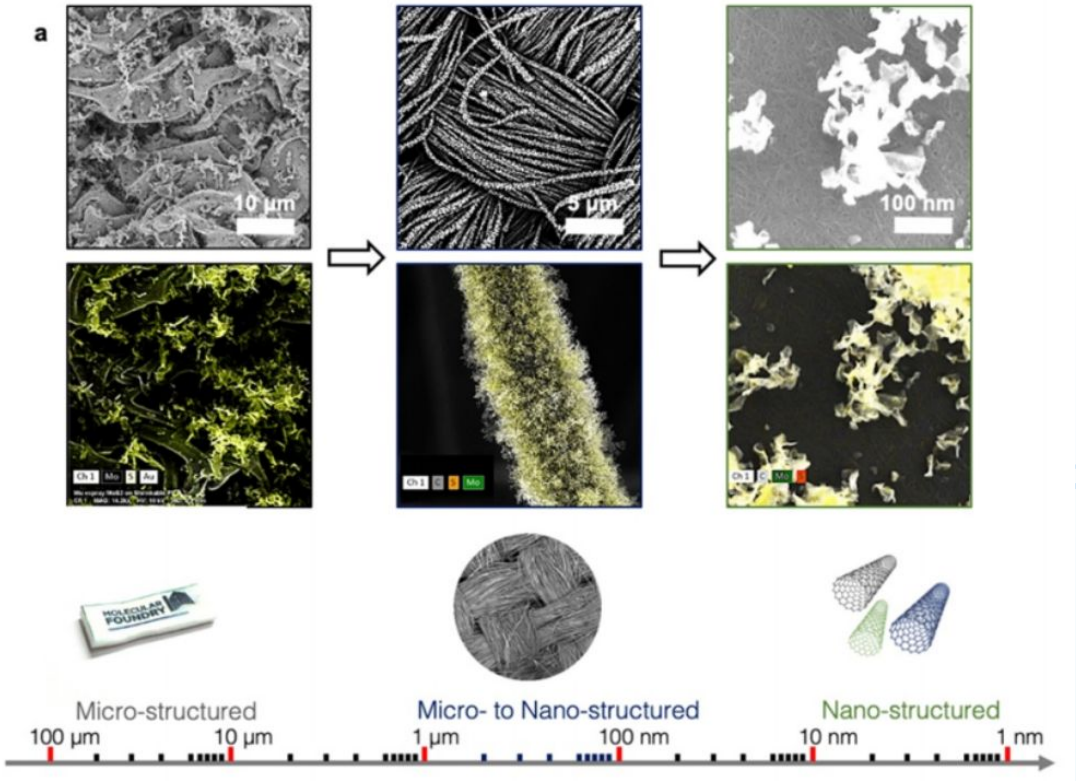
Note: "Y" stands for medium 2 forms a stable spherical droplet on medium 1 and does not induce wetting; whereas "N" indicates that medium 2 preferentially wets and then spreads on medium 1. Specifically, we estimated γ_{12} from the formulation: $\gamma_{12} = \gamma_1 + \gamma_2 - 2(\gamma_1^d\gamma_2^d)^{1/2}$, where γ_1^d and γ_2^d are the dispersion force contributions of the liquid surface tensions. Thus, for H_2O/HOPG, putting $\gamma_1^d = 26$ mJ/m^2 and $\gamma_2^d = 21.8$ mJ/m^2, we calculate $\gamma_{12} \cong 26 + 73 - 2(26 \times 21.8)^{1/2} \cong 51.2$ mJ/m^2, which is very close to the measured value of 48.6 mJ/m^2.

Appendix A-VII: Supplementary table 2

Supplementary Table 2. Comparison of the governing parameters with theoretical prediction and experimental observations on substrates with distinctively different textures, morphologies and surface chemistry, provided that the boundary conditions of $\gamma_{12} > 0$, $S < 0$ and $R < R_i$ are all satisfied.

Substrate	Medium	γ_1 (mJ/m^2)	γ_2 (mJ/m^2)	θ (°)	γ_{12} (mJ/m^2)	S (mJ/m^2)	Spherical droplet	
							Theory	Exp.
SWCNTs	H_2O: IPA	34	73	105	52.9	<0	Y	Y
HOPG	H_2O: IPA	26	73	108	48.6	<0	Y	Y
Polystyrene	H_2O: IPA	40	73	103	56.3	<0	Y	Y
PDMS	H_2O: IPA	19.8	73	96	27.4	<0	Y	Y

Appendix B: Supplementary scheme



- Left panel: microstructured polystyrene;
- Middle panel: micro-to-nano structured CC;
- Right panel: nano-structured SWCNTs, respectively